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## Flexible genetic algorithm for quantum support vector machines

Minh Duc Nguyen<sup>1,3</sup>, Tuan Hai Vu<sup>2,3</sup>, Bin Ho Le<sup>4,5,\*</sup> and Lan Nguyen Tran<sup>1,3</sup><sup>1</sup> Faculty of Physics and Engineering Physics, University of Science, Ho Chi Minh City 70000, Vietnam<sup>2</sup> University of Information Technology, Vietnam National University, Ho Chi Minh City 70000, Vietnam<sup>3</sup> Vietnam National University, Ho Chi Minh City 70000, Vietnam<sup>4</sup> Frontier Research Institute for Interdisciplinary Sciences, Tohoku University, Sendai 980-8578, Japan<sup>5</sup> Department of Applied Physics, Graduate School of Engineering, Tohoku University, Sendai 980-8579, Japan

\* Author to whom any correspondence should be addressed.

E-mail: [binho@fris.tohoku.ac.jp](mailto:binho@fris.tohoku.ac.jp), [haivt@uit.edu.vn](mailto:haivt@uit.edu.vn) and [tnlan@hcmus.edu.vn](mailto:tnlan@hcmus.edu.vn)**Keywords:** quantum computing, machine learning, quantum support vector machine, genetic algorithms, feature map optimization

## Abstract

Quantum support vector machines (QSVM) is one of the most promising frameworks in quantum machine learning, yet their performance depends on the design of the feature map. Conventional approaches rely on fixed quantum circuits, which often fail to generalize across datasets. To address this limitation, we propose genetic algorithms (GA)-QSVM, a hybrid framework that employs GA to automatically optimize feature maps. The proposed method introduces a configurable framework that flexibly defines the evolutionary parameters, enabling the construction of adaptive circuits. Experimental evaluation of datasets, including Digits, Fashion, Wine, and Breast Cancer, demonstrates that GA-QSVMs achieve a comparable accuracy compared to classical support vector machines (SVMs) and standard QSVMs. Although a systematic study of source dataset selection remains an open question, our experiments indicate that GA-QSVM feature maps can be transferred across domains, providing a performance improvement over classical SVM baselines on challenging datasets like Fashion-MNIST. These findings highlight the potential of evolutionary strategies to automate and enhance kernel design for future quantum machine learning applications.

## 1. Introduction

Quantum machine learning (QML) [1, 2] has emerged as a promising field with the potential to offer computational advantages over classical methods. Among various QML algorithms, the quantum support vector machine (QSVM) [3, 4] is particularly notable, as it translates a well-understood classical support vector machine (SVM) into the quantum domain. For the dual problem, a classical SVM has a complexity of  $\mathcal{O}(M^2)$  for an  $M$ -size dataset. In contrast, the quantum dual problem requires  $\mathcal{O}\left(\frac{M^{4.67}}{\epsilon^2}\right)$  quantum circuit evaluations to achieve an  $\epsilon$ -accurate solution [4]. This indicates that the quantum dual approach grows more rapidly with problem size than the classical method. This asymptotic bound, however, is not expected to hold in small-scale simulated regimes. The power of a QSVM lies within its quantum kernel  $K$  between two quantum states  $|\psi(x_i)\rangle$  and  $|\psi(x_j)\rangle$ , which measures the similarity between them. These states are prepared by the quantum feature map  $\phi$  through a unitary  $U_\phi(\theta)$ .

The primary challenge in QSVM is the design of the parameterized circuit used in the feature map. Complex circuits can capture data patterns, but they face two significant issues. First, in high-dimensional feature spaces, the inner product between data vectors  $x_i^T x_j$  concentrates around a constant  $\epsilon$ , which reduces the variability of the kernel matrix, making it nearly uniform and thus difficult to train. Second, overly expressive circuits often lead to poor generalization, a phenomenon analogous to overfitting in classical machine learning [5, 6]. Currently, QSVM parameterized circuits often rely on fixed ansatzes, such as ZZFeatureMap [7], hardware efficient embedding [8, 9] or instantaneous quantum polynomial-style circuits [10, 11], which may not be optimal for a general purpose.

To address the challenge of manual circuit design, previous works used genetic algorithm (GA) to automatically discover effective circuits [12–18]. GA is a powerful metaheuristic for searching in many quantum problems, such as state preparation [13, 14], tomography [15], image processing [16–18], and quantum compilation [19]. In particular, in the QSVM framework, there have been several works employing GA to search for optimal feature maps. In [12], the authors applied GA to search gate-sequence encodings, showing that GA-discovered circuits can outperform hand-designed circuits while revealing a positive relationship between classifier test accuracy and kernel entropy. Pello *et al* [20] demonstrated that the kernel-target alignment is suitable for the direct accuracy evaluation inside an NSGA-II [21] based GA. Those authors further accelerated the fitness evaluation by using an approximation to kernel-target alignment. Gate parameters are optimized with classical optimizers, producing compact maps with improved alignment. Recently, Wang *et al* have studied entanglement explicitly with a multi-objective GA that separates fitness contributions from local and non-local gates [22]. Their work showed that (i) optimal circuits tend to include more non-local gates rather than eliminating them, and (ii) the multi-objective selection yields the Pareto front representing a set of optimal trade-offs where improving accuracy leads to an increase in kernel size.

Existing GA-QSVM approaches still have several limitations. First, their performance often highly depends on the benchmark dataset. As a result, QSVM does not generally outperform classical SVMs on various tasks [12]. Second, previous GA procedures lacked a normalization mechanism to ensure that evolved solutions remain aligned with the original objective [20]. Finally, previous methods lacked well-defined evolutionary strategies to improve fitness in the next generation [22]. To address these limitations, we propose a configurable GA framework that allows the dynamic definition of fitness function, normalizer, generator, and GA operations. Our framework can automatically adjust resources, such as increasing the circuit depth or adding more gates, when GA fails to find an optimal solution. For evaluation, we select four toy datasets that are suitable for simulating QSVM on a classical computer. We then conduct a comparison between classical SVMs, QSVMs with standard feature maps, and our proposed GA-QSVMs. Especially, we also assess the performance of our GA-QSVM with different types of kernels, including Fidelity Quantum Kernel (FQK) and Projected Quantum Kernel (PQK) [6, 23].

The remainder of this paper is organized as follows. Section 2 reviews the related works on QSVM and GA applications in quantum computing. Section 3 presents the background of GA, classical SVM, and quantum kernels. Section 4 describes the proposed GA-QSVM methodology, including the evolutionary design of quantum circuits and metadata configuration. Section 5 provides the evaluation results on benchmark datasets, hyperparameter analysis, and comparison with classical and existing quantum SVMs. Finally, section 6 concludes the paper and discusses potential future research directions.

## 2. Related works

Recently, QSVM has emerged as one of the prominent QML approaches. Several works [3, 5] provided the foundation for applying quantum-enhanced feature spaces to supervised learning tasks. These early studies demonstrated how quantum feature maps could transform classical data into high-dimensional Hilbert spaces, enabling the classification of non-linear patterns that are difficult to capture with classical kernels. Later works have shown that the QSVM performance depends strongly on the dataset [6, 24]. While QSVM and classical SVM perform comparably for simple datasets, QSVM has been found to outperform the classical one for complex datasets thanks to the better representational of quantum feature maps in these cases. Beyond benchmarking, the QSVM framework has been successfully applied to real-world problems. For example, QSVM can solve the vehicle routing problem [25], air pollution prediction [26], DNA sequence similarities [27], and handling data with underlying group structures through covariant kernel formulations [28].

Despite the above-mentioned advances, some major challenges remain. Mapping classical data into quantum states has proven to be non-trivial, with subtleties that affect both trainability and performance [29]. As the same in classical SVM, overfitting is another recurring issue in QSVM [30, 31]. Furthermore, a key challenge in QSVMs lies in the kernel design. Trainable quantum kernels have been proposed to adapt the kernel based on datasets, improving the ability of overall generalization [32, 33]. Complementing this line of work, many automatic circuit design engines have been proposed [34]. Suzuki *et al* [35] analyzed the expressivity of quantum feature maps, providing insights into how different kernels affect the performance of classifiers. Gil-Fuster *et al* showed that one can improve the performance of QSVM by designing new quantum kernels [36, 37]. Recently, the GA-QSVM approach introduced in [12] leveraged GA to optimize kernel structure, representing a step toward automating the

kernel selection problem. This hybrid evolutionary-quantum strategy enhances the accuracy of QSVMs, particularly when dealing with complex datasets [38].

### 3. Background

#### 3.1. GA

GAs are classes of evolutionary optimization techniques inspired by the biological process of natural selection. They operate by maintaining a population that evolves over generations according to a defined fitness function. A population consists of multiple candidates, forming the search space for the algorithm. The structure of each candidate is encoded as a chromosome serving as its genetic representation. Within each chromosome, a gene corresponds to a specific component of the overall solution [39, 40].

The evolution of the population proceeds by repeating selection, crossover, mutation, and replacement operations. In the selection phase, candidates with higher fitness values are more likely to be chosen as parents for the next generation. The crossover operation then combines parts of two or more parent chromosomes to produce offspring, thereby promoting the exchange of better genes. Mutation introduces small random changes to candidate genes, preserving the genetic diversity and helping the algorithm avoid local minima. Finally, the replacement determines which candidates from the current and new generations will survive to become the next population. Through this iterative process, GA gradually refines candidate solutions, converging toward an optimal or near-optimal result. Their ability to search large, complex, and non-convex spaces without relying on gradient makes them particularly effective in optimization [41–43].

Conventional GAs used in existing GA-QSVM frameworks [12, 20, 22] often face limitations due to their static evolutionary mechanisms, leading to a large deviation from the targeted optimization objective. Earlier GA-QSVM implementations also typically lacked a normalization mechanism to ensure that evolved individuals remain consistent with the targeted fitness landscape, resulting in unstable evolution across generations. Additionally, GA frameworks do not have adaptive strategies to enhance fitness progression effectively in generations. To overcome these issues, recent studies have introduced Adaptive GA [44, 45] to dynamically adjust their parameters, such as resources, mutation rate, crossover probability, and selection pressure, based on the current population state. These adaptive mechanisms enhance convergence robustness across various environments.

#### 3.2. Classical SVM

SVM [46, 47] is a popular and effective supervised approach to linear classification. The prediction is made using the function of the inner product of  $x$  and a weight vector  $w \in \mathcal{X} : f^*(x) = \text{sign}(f(x) + b)$ , where  $f(x) = w \cdot x$ , defining a separate hyperplane with orientation and offset controlled by  $w$  and  $b$ , respectively.

To handle non-linearly separable data, a soft margin is employed, and a hinge loss is then traded off with the norm of  $w$  when training SVM models. We can see it as a regularization part to avoid overfitting. For complex and nonlinear problems, data points are mapped from the input space  $\mathcal{X}$  to a higher-dimensional space  $\tilde{\mathcal{X}}$  using a mapping function  $\phi : \mathcal{X} \rightarrow \tilde{\mathcal{X}}$ . In this new space, a linear decision boundary is found. The training process for a soft-margin SVM is defined by the following primal optimization problem:

$$\min_{w, b, \epsilon} \left\{ \frac{1}{2} \|w\|_2^2 + C \sum_{i=1}^n \epsilon_i \right\} \quad (1)$$

subject to:

$$\epsilon_i \geq 0, \text{ and } y_i (w \cdot \phi(x_i) + b) \geq 1 - \epsilon_i \quad \forall i = 1, \dots, n, \quad (2)$$

where the parameter  $C > 0$  determines the trade-off between increasing the margin size and ensuring that the  $x_i$  lie on the correct side of the margin.

The kernel trick is used to avoid explicitly computing the mapping  $\phi(x_i)$ . By using a kernel function,  $K(x_i, x_j)$ , we can calculate the inner product of vectors in  $\tilde{\mathcal{X}}$  directly from the original input vectors. This allows the problem to be solved using its dual formulation, expressed as:

$$\max_{\beta} \left\{ L(\beta) = \sum_{i=1}^n \beta_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \beta_i \beta_j y_i y_j K(x_i, x_j) \right\}, \quad (3)$$

**Algorithm 1.** QSVM  $\mathcal{F}$ .

---

**Require:** Dataset  $D$ , quantum circuit  $\mathbf{g}$   
 1: Construct quantum kernel  $K$  with  $\mathbf{g}$  as feature map  
 2: Construct  $\{x_{\text{train}}, y_{\text{train}}, x_{\text{test}}, y_{\text{test}}\}$  from  $D$ .  
 3: Train the classifier  $\mathbf{C}_K$  with  $\{x_{\text{train}}, y_{\text{train}}\}$   
 4:  $y_{\text{pred}} \leftarrow \mathbf{C}_K(x_{\text{test}})$   
 5: **return** Accuracy between  $\{y_{\text{test}}, y_{\text{pred}}\}$

---

subject to:

$$\sum_{i=1}^n \beta_i y_i = 0, \text{ and } 0 \leq \beta_i \leq C \forall i = 1, \dots, n. \quad (4)$$

Classical SVMs fundamentally rely on user-chosen kernels whose inductive bias may not capture higher-order, non-local correlations in the data. In practice, performance depends heavily on heuristic kernel selection and bandwidth tuning; poorly tuned kernels either overfit (small bandwidth) or “wash out” structure (large bandwidth), and hyperparameters interact in nontrivial ways [48]. Moreover, kernel methods scale at least quadratically in sample size due to the  $N \times N$  Gram matrix, creating time/memory bottlenecks for large  $N$  and complicating cross-validation and model selection. Even sophisticated classical kernels or twin-SVM variants cannot fully avoid these scaling limits [49].

### 3.3. QSVM

QSVMs are quantum versions of classical SVMs. As defined in algorithm 1, they leverage quantum principles to handle high-dimensional data. QSVMs define kernels via state fidelity in exponentially large Hilbert spaces using data-encoding circuits. Carefully designed circuits can embed classically hard-to-compute similarities, potentially improving generalization for certain tasks [48]. In QSVM, the datapoint  $x_i$  is encoded into a quantum state  $|\psi(x_i)\rangle$ . This mapping is achieved through a quantum feature map  $U(\theta, x)$  applied to an initial state  $|0\rangle$ :

$$|\psi(x)\rangle = U(\theta, x)|0\rangle. \quad (5)$$

The widely-used QSVM kernel is FQK  $K$ , defined as the fidelity between two quantum states encoded by the circuit  $\mathbf{g}$ , denoted as  $K_{\mathbf{g}}^{\text{FQK}}(x_i, x_j)$ :

$$K_{\mathbf{g}}^{\text{FQK}}(x_i, x_j) = |\langle \psi(x_i) | \psi(x_j) \rangle|^2. \quad (6)$$

Each input  $x_i$  is then fed into classifier  $\mathbf{C}_K$ , which returns the corresponding label  $y_i$ .

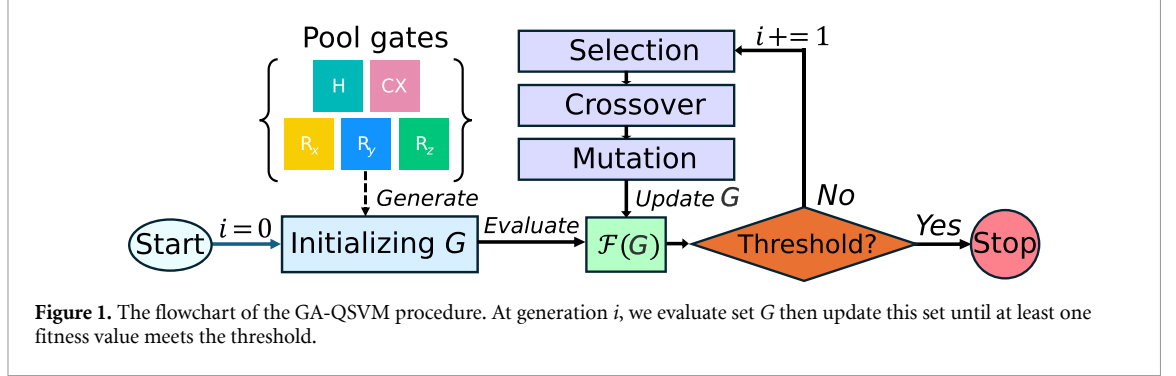
To overcome the drawback of standard FQK suffering from small geometric differences when the effective dimension  $d$  is large, PQK has been recently developed by Huang *et al* [6]. PQK projects quantum states into approximate classical representations, such as reduced physical observables or classical shadows, enabling better generalization in low-dimensional classical spaces. The simple form of PQK measures the one-particle reduced density matrix (1-RDM) for each qubit:  $\rho_k(x_i) = \text{Tr}_{j \neq k}[\rho(x_i)]$  and is defined as

$$K_{\mathbf{g}}^{\text{PQK}}(x_i, x_j) = \exp \left( -\gamma \sum_k \|\rho_k(x_i) - \rho_k(x_j)\|_F^2 \right), \quad (7)$$

which is known as the Frobenius norm between two 1-RDM. PQK can be efficiently computed via classical shadows, providing a practical route for quantum-enhanced learning with fewer measurements. Recent works have shown that quantum kernels are efficient in multiclass classification when quantum circuits and their hyperparameters are chosen properly [8, 50, 51]. In particular, QSVMs can bypass classical SVM limitations when (i) the quantum feature map targets data structures inapplicable to classical kernels, and (ii) hyperparameters are carefully managed [51]. The GA approach with dynamically designing quantum circuits for the kernel is expected to solve the two above-mentioned problems.

**Table 1.** GA-QSVM terminology mapping.

| GA         | GA-QSVM                                    | Description                              |
|------------|--------------------------------------------|------------------------------------------|
| Gene       | Quantum gate $g$                           | Unitary operator samples from Pool gates |
| Candidate  | Parameterized quantum circuit $\mathbf{g}$ | A set of $g$                             |
| Population | Generation $G^{(i)}$                       | A set of $\mathbf{g}$                    |
| Fitness    | Fitness                                    | Performance metric                       |

**Algorithm 2.** GA for QSVM task.

---

**Require:** Fitness function  $\mathcal{F}$ , metadata  $\mathcal{M}$

- 1:  $G^{(0)} \leftarrow \text{Generator}(\mathcal{M})$
- 2: **for**  $i$  in  $[0 \dots n_{\text{generation}}]$  **do**
- 3:    $F \leftarrow [\mathcal{F}(\mathbf{g}) \text{ for } \mathbf{g} \text{ in } G^{(i)}]$
- 4:   **if**  $\exists f \in F, f < \mathcal{M}\tau$  **then**
- 5:     **return**  $G^{(i)}$
- 6:   **end if**
- 7:    $\{G_{\text{father}}^{(i)}, G_{\text{mother}}^{(i)}\} \leftarrow \text{Selection}(G^{(i)}, \text{Metadata})$
- 8:    $G^{(i+1)} \leftarrow \text{Mutation}(\text{Crossover}(\{G_{\text{father}}^{(i)}, G_{\text{mother}}^{(i)}\}, \dots), \dots)$
- 9: **end for**
- 10: **return**  $G^{(i)}$

---

## 4. Methodology

Our approach combines GA and QSVM into a hybrid workflow as a two-level optimizer. QSVM serves as a low-level optimizer to find an optimal hyperplane, whereas GA is the high-level optimizer to find the best circuit, which strongly affects the QSVM optimization. Table 1 provides a terminology mapping between conventional GA concepts and their interpretations in the proposed GA-QSVM framework.

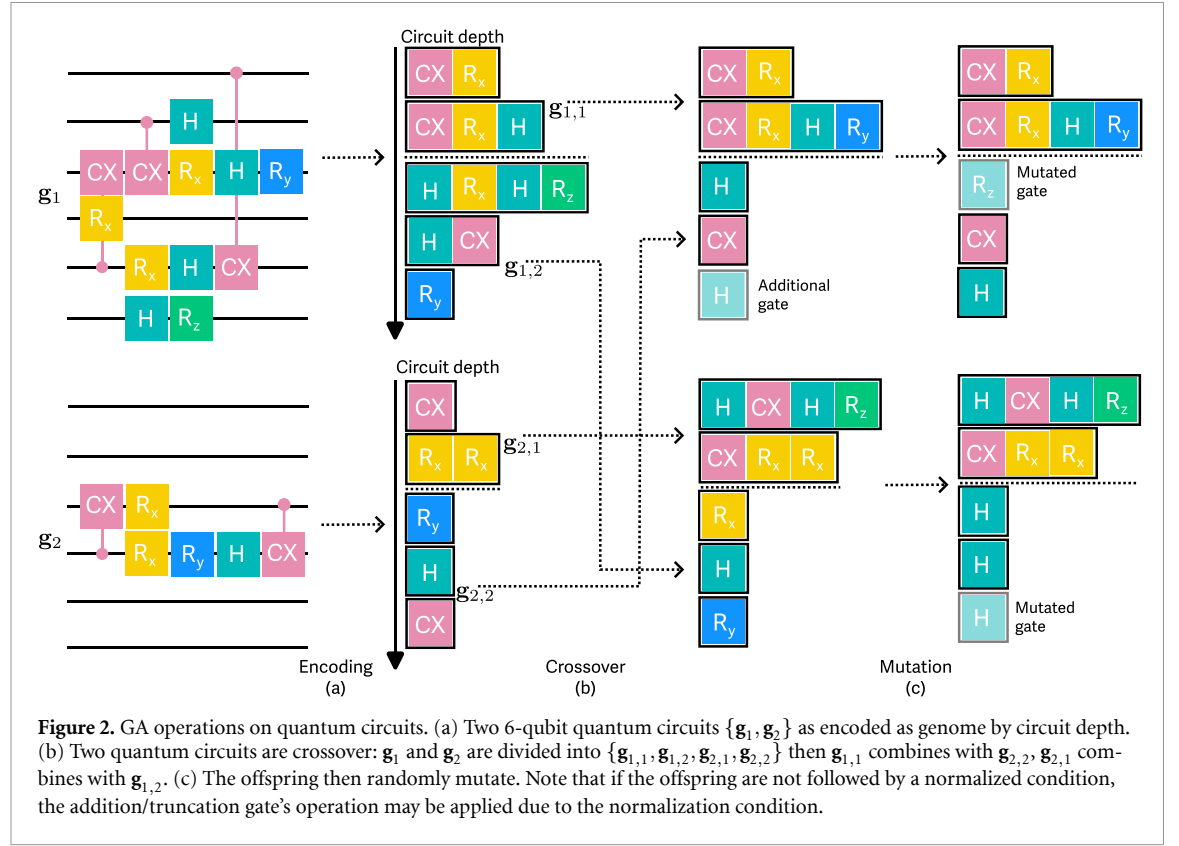
### 4.1. GA-QSVM procedure

Our GA-QSVM procedure is presented in figure 1. Each candidate in the population represents a circuit composed of a sequence of gates from a pool. Notice that previous works included  $R_X$ ,  $R_Z$ , and CX gates restricted to 90-degree ( $\sqrt{X}$ ), but lacked the Y-axis rotation gate [20]. In this work, to enhance the flexibility, we involve  $R_X(\theta)$ ,  $R_Y(\theta)$ ,  $R_Z(\theta)$ , H, CX gates.

The corresponding fitness  $\mathcal{F}(\dots)$  of each circuit is determined by its performance for a given dataset  $D$  that is separated into  $\{D_{\text{train}}, D_{\text{test}}\}$ . As given in algorithm 2, for each circuit in the population, we perform QSVM using the corresponding quantum kernel. QSVM is trained on  $D_{\text{train}}$  and its classification accuracy is measured on  $D_{\text{test}}$ , this score serves as the fitness:

$$\text{Accuracy} = \frac{1}{|y_{\text{test}}|} \sum_{j=1}^{|y_{\text{test}}|} \mathbb{I}(\hat{y}_{\text{test}_j} = y_{\text{test}_j}), \quad (8)$$

where  $\mathbb{I}(\cdot)$  is the indicator function, which is 1 if the predicted label  $\hat{y}$  matches the true label  $y$  and 0 otherwise.



At the end of each generation, the circuits are ranked by their fitness. The evolution then proceeds with three steps: selection, crossover, and mutation, as depicted in figure 2. We employ the elitist selection strategy, in which the best-performing circuits of the population are chosen to be parents for the next generation. The selected circuits are then divided into a father set and a mother set. The number of eliminated circuits is set to  $n_{\text{circuit}}/2$  by default, meaning that the population is kept over generation if the initial population is divisible by 4.

$$\{G_{\text{father}}^{(i)}, G_{\text{mother}}^{(i)}\} \leftarrow \text{Selection.Elitist}(G^{(i)}, \mathcal{F}, n_{\text{circuit}}/2). \quad (9)$$

These selected circuits undergo crossover to create a new generation of offspring. We stress that, unlike previous GA-QSVM frameworks [20], our approach forces the offspring to be normalized under several conditions that balance the expressivity and trainability of quantum circuits. For example, (1) circuits should be shortened if their depth is longer than a pre-defined depth  $d$ , and (2) CX gates must be added if the number of CX gates is less than the number of required CX gates.

$$G^{(i+1)} \leftarrow \text{Crossover.OnePoint}\left(\{G_{\text{father}}^{(i)}, G_{\text{mother}}^{(i)}\}, \text{Normalizer.Depth}(d), \dots\right). \quad (10)$$

We use the bit-flip mutation with a mutation rate  $p$ . For each gate  $g$ , there is a  $p\%$  chance that  $g$  changes to a different gate belonging to the pool gates. The value  $p$  is thus crucial to find an optimal circuit. If  $p$  is low, GA-QSVM will get stuck at local minima; otherwise, the evolution process can be unstable. The mutated circuits also need to be normalized following the normalization condition:

$$G^{(i+1)} \leftarrow \text{Mutation.BitFlip}\left(G^{(i+1)}, \text{Normalizer.Depth}(d), \dots\right). \quad (11)$$

To avoid implicit test-set leakage, we adopt a two-stage evaluation protocol. In the first stage, the GA evolves circuits on a dedicated train/validation split that is held fixed throughout the evolutionary search. In the second stage, the best evolved circuit is evaluated on independent train/test splits using  $k$ -fold cross-validation with different random seeds, ensuring that no data seen during GA optimization appears in the final evaluation.



**Table 2.** Dataset properties.

| Dataset       | References | #Instances | #Classes | #Features |
|---------------|------------|------------|----------|-----------|
| Digits        | [56]       | 5620       | 10       | 64        |
| Fashion       | [57]       | 70 000     | 10       | 784       |
| Wine          | [58]       | 178        | 3        | 13        |
| Breast Cancer | [59]       | 592        | 2        | 30        |

#### 4.2. Metadata for GA-QSVM

In the proposed GA-QSVM framework, *metadata* is defined as a tuple of hyperparameters that configure both the quantum feature map and the GA process. Formally, we denote the metadata as

$$\mathcal{M} = (\tau, n, n_{R_x}, n_{R_y}, n_{R_z}, d, n_{\text{circuit}}, n_{\text{generation}}, p), \quad (12)$$

where  $\tau$  is the threshold,  $n$  is the number of qubits,  $(n_{R_x}, n_{R_y}, n_{R_z})$  represent the allocation of single-qubit rotation gates along the  $x$ ,  $y$ , and  $z$  axes,  $d$  denotes the circuit depth,  $n_{\text{circuit}}$  is the number of circuits in the population,  $n_{\text{generation}}$  is the number of generations, and  $p$  is the mutation probability.

The first subset of parameters  $(n, n_{R_x}, n_{R_y}, n_{R_z}, d)$  governs the expressibility of the quantum circuit, while the second subset  $(n_{\text{circuit}}, n_{\text{generation}}, p)$  regulates the behavior of the GA by controlling the balance between exploration and exploitation. By encapsulating these parameters in  $\mathcal{M}$ , the framework ensures reproducibility, enables systematic benchmarking across hyperparameter configurations. In particular, the circuit depth  $d$  determines the representational capacity of the evolved quantum feature map by controlling the number of sequential operations and entangling interactions. Larger depths generally allow the circuit to capture more complex feature correlations, although excessively deep circuits may increase optimization difficulty and computational cost. The mutation probability  $p$  regulates the exploration behavior of the GA. Small mutation rates may cause premature convergence to local optima, whereas overly large mutation rates can disrupt promising circuit structures and destabilize evolution. Similarly, the population size  $n_{\text{circuit}}$  affects the diversity of candidate solutions, while the number of CX gates influences the entanglement capability and nonlinearity of the quantum feature map.

#### 4.3. Computational cost

While GA-QSVM successfully automates kernel design, the computational cost of the evolutionary search is a notable trade-off. For  $n_{\text{circuit}}$  circuits evolving over  $n_{\text{generation}}$  generations on a dataset of size  $M$ , the total number of quantum circuit evaluations scales as  $\mathcal{O}(n_{\text{generation}} \times n_{\text{circuit}} \times M^2)$ . For example, with  $M = 100$ ,  $n_{\text{circuit}} = 20$ , and  $n_{\text{generation}} = 200$ , the search requires approximately  $4 \times 10^7$  quantum evaluations. The parameters  $n$  and  $d$  influence the computational cost of each quantum circuit evaluation exponentially and polynomially on run time, respectively. Finally, the run time complexity is  $\mathcal{O}(n_{\text{generation}} \times n_{\text{circuit}} \times M^2 \times 2^n \times d)$ . Although we mitigated this in our simulations by reducing the number of QSVM training iterations during fitness evaluation, the wall-clock time remains significantly higher than that of fixed-ansatz QSVMs. This high overhead characterizes GA-QSVM primarily as an “offline” discovery tool rather than a frequent training framework. Once identified, the optimized circuit can be used “online” for inference or transfer learning with a cost equivalent to standard QSVMs.

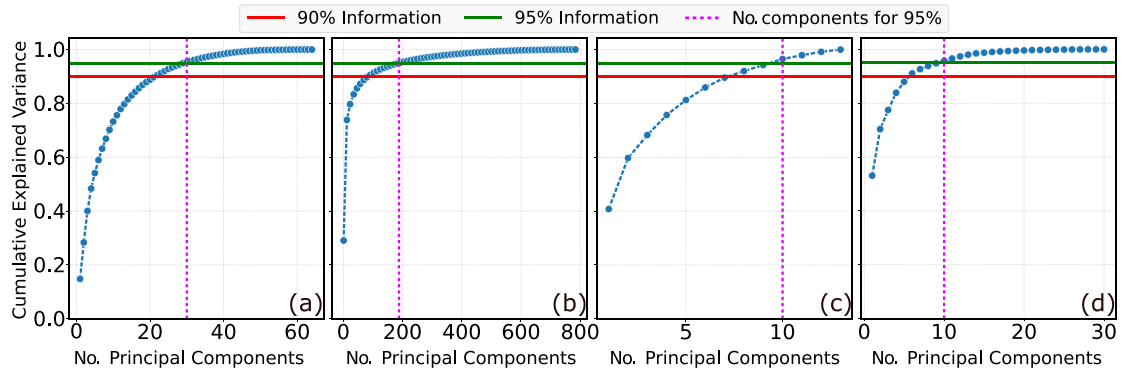
## 5. Numerical results

### 5.1. Dataset preparation

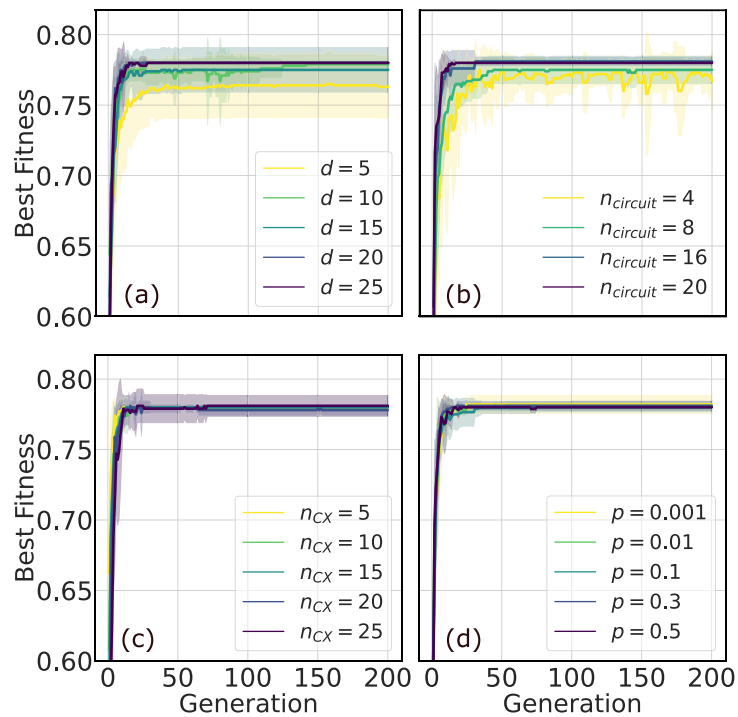
We have performed QSVM with two types of quantum kernels, including PQR and FQR using Qiskit 1.3.1 [52] and Squearn 0.8.4 [53], respectively. Regarding datasets, we used Scikit-learn 1.6.1 [54] for Wine, Digits, and Breast Cancer datasets and Tensorflow 2.16.2 [55] for the Fashion MNIST dataset. All calculations were performed on an Intel Core i9-10920X CPU at 3.50 GHz and 120 GB of RAM. For comparison, we have used a radial basis (RBF) kernel for classical SVM, and Z/ZZ feature map with FQR and PQR for conventional QSVM. The default configuration  $\mathcal{M}_0$  is  $\tau = 0.99, n = 10, d = 5, n_{\text{circuit}} = 16, n_{\text{generation}} = 200, p = 0.1$ .

We use four datasets for the classification task, including: **Digits**, **Fashion**, **Wine**, and **Breast Cancer** as described in table 2. The maximum number of training items and testing items is 100 and 100 for all experiments, respectively. This restriction arises from the  $M \times M$  kernel matrix construction required at each fitness evaluation. **Fashion** was excluded from the initial GA-QSVM training phase due to the high number of features (#Features). For the other three datasets, principal component analysis (PCA)





**Figure 3.** Cumulative explained variance of high complexity ((a) **Digits**, (b) **Fashion**), and Low complexity ((c) **Wine**, and (d) **Breast Cancer**) dataset after using PCA. The number of components for 95% cumulative explained variance for four datasets is 30, 200, 10, and 10.



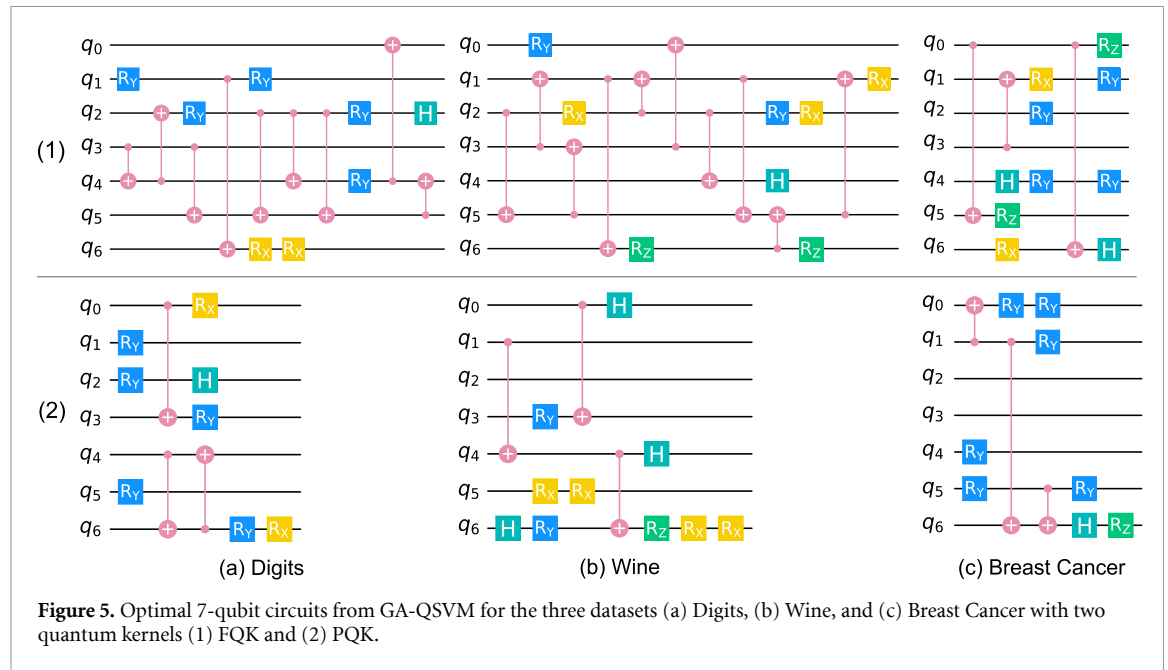
**Figure 4.** We conduct the survey on hyperparameters by benchmarking on **Digits** dataset, with fixed  $n = 5$ ,  $n_{\text{circuit}} = 16$ ,  $p = 0.1$ ,  $d = 5n$ ,  $n_{\text{CX}} = 2n$  and different configuration. (a) different  $d$ , (b) different  $n_{\text{circuit}}$ , (c) different  $n_{\text{CX}}$  and (d) different  $p$ . Note that in GA-QSVM, we run the fitness function (QSVM) with fewer iterations to save the whole optimization time.

was applied to reduce #Features. To prevent data leakage, PCA was fitted exclusively on the training set and subsequently applied to the test set without refitting. Note that the number of qubits must be equal to #Features because we use the angle encoding method. Preliminary results are shown in figure 3, the number of principal components needed to retain a significant amount of information varies greatly by dataset. The **Wine** and **Breast Cancer** datasets are less complex, requiring only 10 components for 95% information, while the **Digits** and **Fashion** datasets, especially the latter, require a much higher number of components.

We recognize that this small-data regime is distinct from large-scale deep learning benchmarks. In this regime, quantum kernels are uniquely positioned to leverage complex, non-linear feature maps that can outperform standard classical kernels which may suffer from high bias when training data is scarce.

## 5.2. Optimal circuit searching

As standard GA, we measure the best fitness versus generation. As an example, we plot the searching process for the digit dataset in figure 4. The best fitness starts from a low value, becomes better through generations, and reaches an upper bound limit that comes from the resources we restrict and the dataset.



**Figure 5.** Optimal 7-qubit circuits from GA-QSVM for the three datasets (a) Digits, (b) Wine, and (c) Breast Cancer with two quantum kernels (1) FQK and (2) PQK.

**Table 3.** Classification accuracy over 10 independent GA runs in figure 4(a) with  $d = 25$ .

| Generation | Accuracy (Mean $\pm$ Std) |
|------------|---------------------------|
| 10         | $0.775 \pm 0.005$         |
| 50         | $0.781 \pm 0.003$         |
| 100        | $0.782 \pm 0.002$         |
| 200        | $0.783 \pm 0.002$         |

We investigate how hyperparameters affect the best fitness. The benchmark results show the effect of different configurations:  $n_{CX} \in [5, 10, 15, 20, 25]$ ,  $p \in [0.001, 0.01, 0.1, 0.3, 0.5]$ ,  $d \in [5, 10, 15, 20, 25]$ ,  $n_{circuit} \in [4, 8, 16, 20]$ .

As shown in figure 4(a), the circuit depth  $d$  has a noticeable effect on the performance; deeper circuits, such as  $d = 20$  and  $d = 25$ , lead to a higher and more stable fitness. The size of the population  $n_{circuit}$ , shown in figure 4(b), has a significant impact on performance and convergence. Larger populations, such as  $n_{circuit} = 16$  and  $n_{circuit} = 20$ , converge more quickly and achieve a more stable final fitness with less variance. As we can see in figures 4(c) and (d), the best fitness for different  $n_{CX}$  and  $p$  values are tightly close to each other with the peak fitness around 0.79. This suggests that the fitness landscape is nearly flat w.r.t  $n_{CX}$  and  $p$ , we can eliminate hyperparameter tuning for  $n_{CX}$  and  $p$ . Therefore, GA-QSVM's performance is dominated more by the structural choice of gates than by the quantity of CX gates. Overall, as the depth and the population size increase, more resources are included in the GA process, leading to better circuit searching.

Figure 5 illustrates the optimal 7-qubit circuits selected by the GA-QSVM framework. Each figures (a)–(c) corresponds respectively to the Digits, Wine, and Breast Cancer datasets, while rows (1) and (2) show the best circuits produced under the FQK and PQK kernels. For all datasets, GA yields shallow circuits combining  $\{R_X, R_Y, R_Z, H, CX\}$ . Notably, the FQK-optimized circuits tend to add more CX gates, whereas PQK-optimized circuits often emphasize balance between local and non-local gates.

As shown in table 3, the variation across runs is relatively small, with the mean accuracy improving from  $0.775 \pm 0.005$  after 10 generations to  $0.783 \pm 0.002$  after 200 generations. The increasing mean and decreasing standard deviation indicate that the evolutionary process converges toward stable solutions despite the stochastic nature of the GA.

### 5.3. GA-QSVM performance

To ensure unbiased evaluation, the GA optimization and final performance reporting are conducted on strictly separated data splits. Specifically, GA fitness is computed on a fixed train/validation split, while final accuracies reported in figure 6 are obtained via  $k$ -fold cross-validation on independent splits with different random seeds.

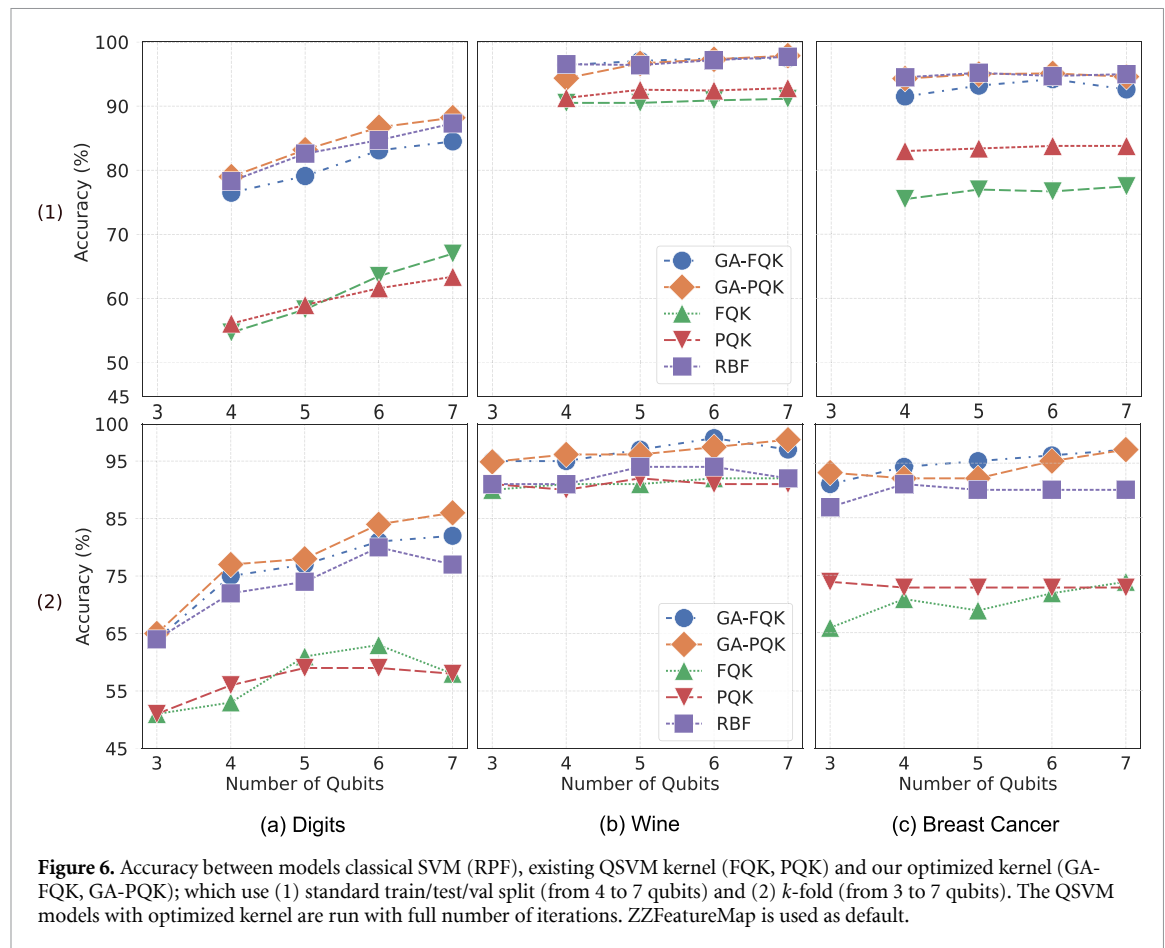


Figure 6 compares the accuracy of different methods (GA-FQK, GA-PQK, FQK, PQK, RBF) for three datasets using the standard train/test/val split (figure (1)) and  $k$ -fold cross-validation (figure (2)). In  $k$ -fold cross-validation, the QSVM is trained  $k$  times through  $k$  train–test splits, then the results are averaged to obtain the final accuracy [60]. The performance of the standard train/test/val method is not stable because it depends on how the test set is split. In contrast,  $k$ -fold cross-validation yields a more robust estimate of the model’s generalization performance [61].

We can see that the conventional QSVM (FQK/PQK) with a fixed ansatz shows the worst accuracy for all cases. For the standard train/test/val method, the performance of GA-FQK/GA-PQK and RBF is comparable. However, the performance of GA-QSVM with both FQK and PQK shows a consistent improvement over the classical RBF kernel. (figure (2)). In general, the comparison highlights that our proposed methods maintain competitive or superior performance under the standard evaluation, and both quantum and classical SVMs achieve near-saturation accuracy.

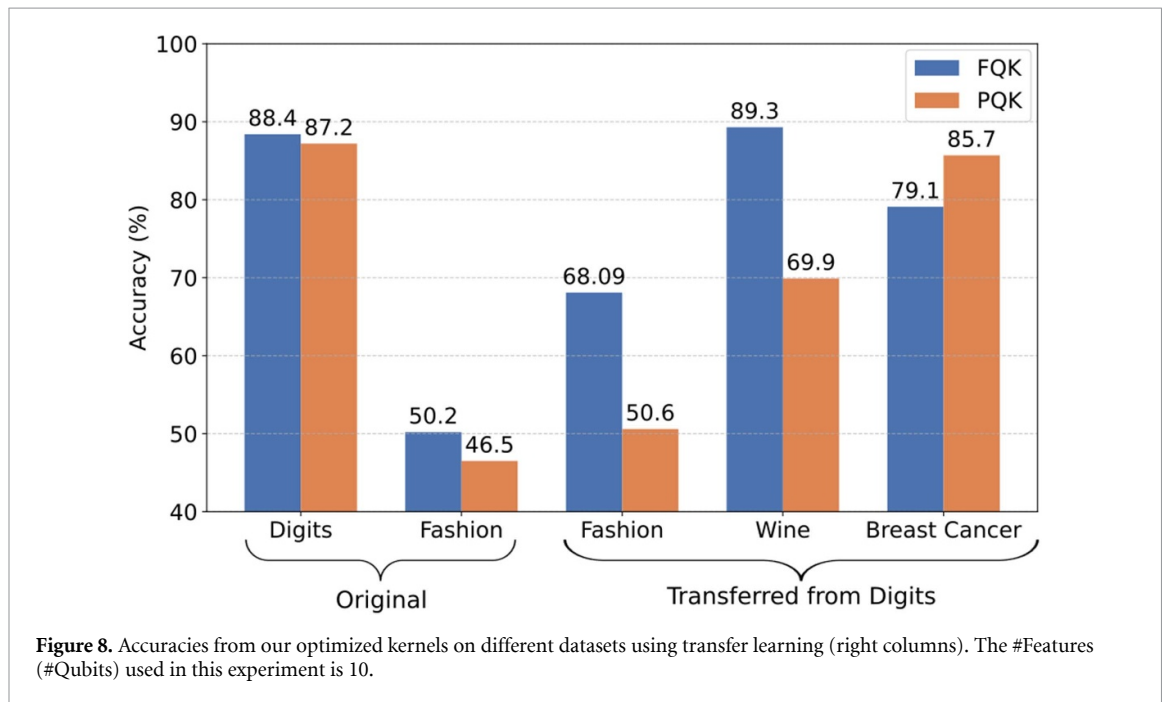
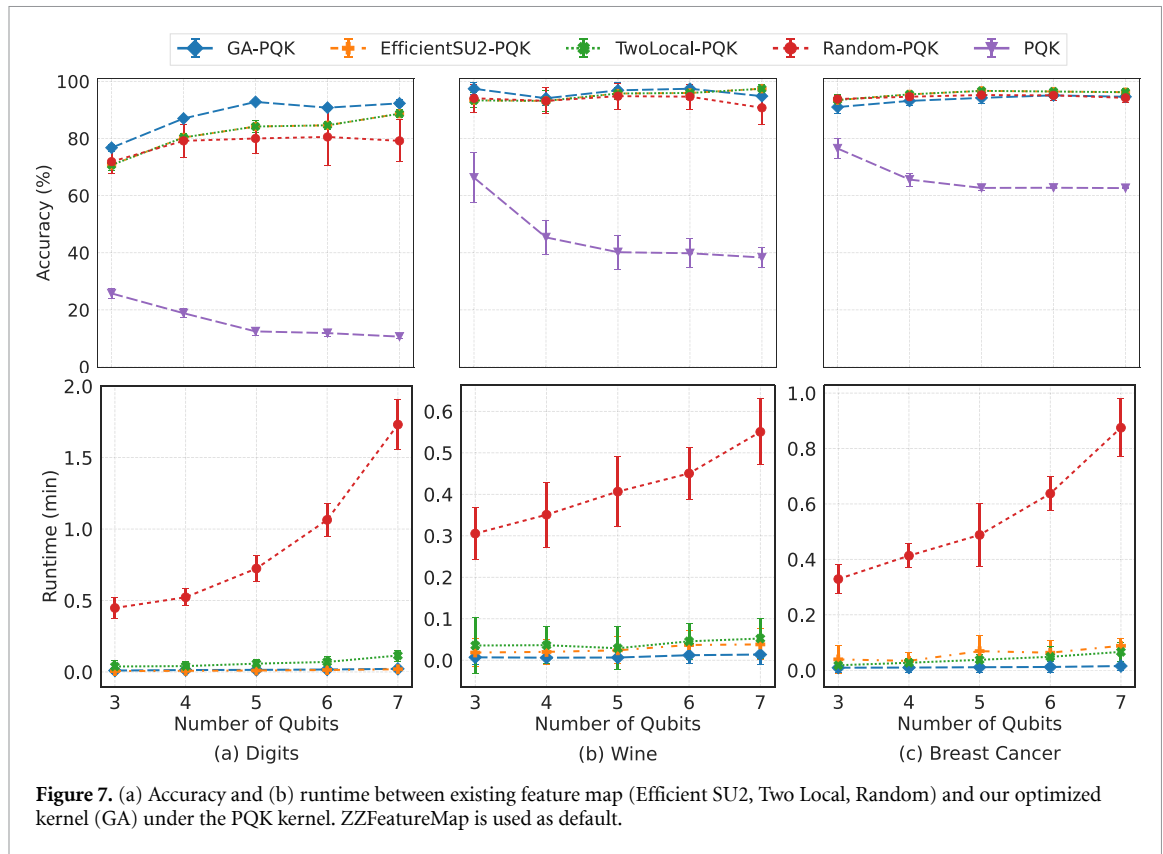
We experiment with more existing feature map and random feature map circuits in figure 7. The results show that our feature map still outperforms existing ones in almost every case, in both accuracy and runtime.

#### 5.4. Transfer learning

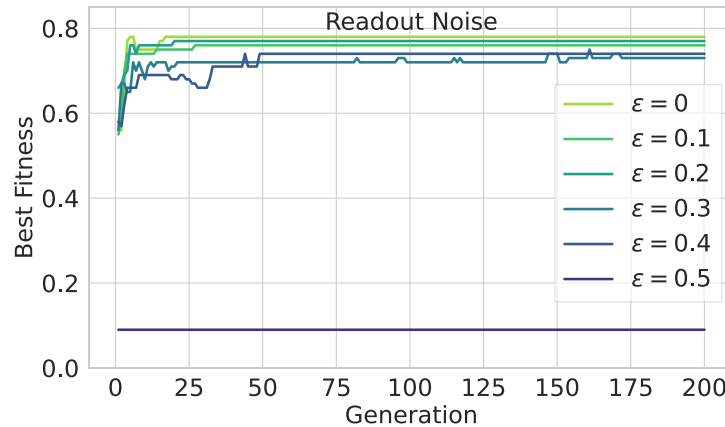
To investigate the generalizability of our GA-discovered kernels, we utilize a High-to-High and High-to-Low transfer strategy. We select Digits (high complexity) as the source dataset, as the GA-QSVM is forced to evolve highly expressive circuits to handle its intricate feature correlations.

This expressive structure is then expected to transfer to other datasets. Transfer learning aims to reduce the number of trials for GA-QSVM on other datasets. While a systematic study across all source datasets is beyond the scope of this work due to computational constraints, this choice provides a strong baseline for evaluating cross-domain transferability.

Figure 8 summarizes the performance of transfer learning across different target datasets when using **Digits** as the base dataset. To enhance the expressivity of the transferred circuit, we have used 10 features corresponding to 10 qubits. The first two columns are results from the GA-QSVM original training for **Digits** and **Fashion**, whereas the last three columns represent the accuracy of QSVM for **Fashion**, **Wine**, and **Breast Cancer** using the ansatz transferred from **Digits**.



**Fashion** is a complex dataset that the classical SVM can only achieve an accuracy of 65.7%, and QSVM/PQK with the standard Z-Feature Map (with  $n = 10$ ) has a very low accuracy of 22.9%. Also, we can see that the **Fashion** dataset is so complex that direct GA-QSVM training achieves relatively low accuracies (only 50.2% and 46.5% for FQK and PQK, respectively). However, when the ansatz transferred from **Digits** is used for **Fashion**, the accuracy is significantly improved. Especially, the FQK accuracy is up to 68.1%, higher than the classical SVM (65.7%). This improvement may arise from the ansatz learned on **Digits**, capturing richer feature correlations and offering a more favorable inductive bias than training on **Fashion** alone. For **Wine** and **Breast Cancer**, the transfer learning can also achieve



**Figure 9.** Readout noise benchmarking under configurations like figure 4 with  $\epsilon \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$  and  $d = 10$ .

reasonable accuracies. The comparison also reveals that FQK overall outperforms PQK in transfer learning, except for the **Breast Cancer** where PQK shows superior performance (85.7 vs 79.1). In general, using transfer learning can help to reduce the computational demands of GA training. However, one drawback is that circuits optimized for **Digits** and **Fashion** tend to exhibit larger depth. Still, an open question is how to choose a proper base dataset to get an expressive ansatz for various datasets.

### 5.5. Noisy effect

To assess the practicality of the proposed approach on near-term quantum hardware, we performed noise simulations incorporating realistic readout errors. Readout noise is adopted as a representative noise model because the physical error sources in NISQ devices, including gate errors, decoherence ( $T_1/T_2$  relaxation), crosstalk, and state preparation imperfections, ultimately manifest as incorrect measurement outcomes [62, 63]. The results are shown in figure 9. As expected, all quantum-kernel methods experience a reduction in classification accuracy under noisy conditions. Nevertheless, the best fitness remains acceptable ( $> 0.7$ ) under  $\epsilon \leq 0.4$ , suggesting that the evolutionary search tends to favor circuit structures with balanced expressivity and noise resilience.

## 6. Conclusion

By allowing flexible metadata configuration and adaptive search, GA-QSVM overcomes the limitations of manually designed circuits and improves kernel expressivity. Numerical results on various datasets demonstrate that GA-QSVM achieves an accuracy superior to standard QSVM with a fixed quantum feature map, especially for complex data. Although our transfer learning results show that these kernels are not yet universal, the consistent improvement over classical baselines, even if modest.

The GA-QSVM approach also presents limitations. The optimization process is still computationally expensive due to repeated kernel evaluations, and the stochastic nature of GA can lead to instabilities across runs. Additionally, as the number of qubits increases, circuit search and evaluation becomes much more difficult, restricting scalability on near-term devices. Future work will focus on extending GA-QSVM to multi-objective optimization, enabling simultaneous control over accuracy, circuit depth, and entanglement cost. One important direction is scaling GA-QSVM to larger datasets, such as using kernel approximations, to reduce the  $M^2$  evaluation overhead and validate our findings in the high-data regime. These directions may further enhance the practicality of evolutionary quantum kernel design for real-world machine learning tasks.

Despite these results, several limitations remain. Exact quantum simulation scales exponentially with the number of qubits  $n$ , restricting the framework to small-scale problems. The angle encoding strategy requires PCA-based dimensionality reduction, which may discard task-relevant information. Moreover, the GA search remains sensitive to random initialization, affecting reproducibility.

We note that the current results should be interpreted as a proof-of-concept for the NISQ-era regime. Scaling GA-QSVM to larger datasets will require moving beyond exact simulation toward kernel approximation methods or hardware-native execution to reduce the  $M^2$  evaluation overhead. Addressing these challenges through hardware-aware circuit design and alternative encoding schemes will be the subject of future work.

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## Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://github.com/vutuanhai237/GA-QSVM> [64].

## Author contributions

Minh Duc Nguyen

Investigation (equal), Software (equal), Validation (equal), Visualization (equal), Writing – original draft (equal), Writing – review & editing (equal)

Tuan Hai Vu

Formal analysis (equal), Investigation (equal), Methodology (equal), Software (equal), Supervision (equal), Validation (equal), Writing – original draft (equal), Writing – review & editing (equal)

Bin Ho Le  0000-0002-8816-4450

Funding acquisition (equal), Supervision (equal), Writing – original draft (equal), Writing – review & editing (equal)

Lan Nguyen Tran

Formal analysis (equal), Funding acquisition (equal), Methodology (equal), Project administration (equal), Supervision (equal), Writing – original draft (equal), Writing – review & editing (equal)

## References

- [1] Schuld M and Killoran N 2019 Quantum machine learning in feature Hilbert spaces *Phys. Rev. Lett.* **122** 040504
- [2] Schuld M, Sinayskiy I and Petruccione F 2015 An introduction to quantum machine learning *Contemp. Phys.* **56** 172–85
- [3] Rebentrost P, Mohseni M and Lloyd S 2014 Quantum support vector machine for big data classification *Phys. Rev. Lett.* **113** 130503
- [4] Gentinetta G, Thomsen A, Sutter D and Woerner S 2024 The complexity of quantum support vector machines *Quantum* **8** 1225
- [5] Havlíček V, Córcoles A D, Temme K, Harrow A W, Kandala A, Chow J M and Gambetta J M 2019 Supervised learning with quantum-enhanced feature spaces *Nature* **567** 209–12
- [6] Huang H-Y, Broughton M, Mohseni M, Babbush R, Boixo S, Neven H and McClean J R 2021 Power of data in quantum machine learning *Nat. Commun.* **12** 2631
- [7] Schuld M, Bocharov A, Svore K M and Wiebe N 2020 Circuit-centric quantum classifiers *Phys. Rev. A* **101** 032308
- [8] Thanasi P, Wang S, Cerezo M and Holmes Z 2024 Exponential concentration in quantum kernel methods *Nat. Commun.* **15** 5200
- [9] Leng J, Li J, Peng Y and Wu X 2025 Expanding hardware-efficiently manipulable Hilbert space via hamiltonian embedding *Quantum* **9** 1857
- [10] Armengol E and Bowles J 2026 Iqopt: fast optimization of instantaneous quantum polynomial circuits in jax (arXiv:2501.04776)
- [11] Hangleiter D, Kalinowski M, Bluvstein D, Cain M, Maskara N, Gao X, Kubica A, Lukin M D and Gullans M J 2025 Fault-tolerant compiling of classically hard instantaneous quantum polynomial circuits on hypercubes *PRX Quantum* **6** 020338
- [12] Creevey F M, Heredge J A, Sevier M E, and Hollenberg L C L 2026 Kernel alignment for quantum support vector machines using genetic algorithms (arXiv:2312.01562)
- [13] Creevey F M, Hill C D and Hollenberg L C L 2023 Gasp: a genetic algorithm for state preparation on quantum computers *Sci. Rep.* **13** 11956
- [14] Fomichev S *et al* 2024 Initial state preparation for quantum chemistry on quantum computers *PRX Quantum* **5** 040339
- [15] Linh H L D, Hai V T and Ho L B 2026 Advancing quantum process tomography through quantum compilation *Adv. Quantum Technol.* **9** e00494
- [16] Altares-López S, García-Ripoll J J and Ribeiro A 2024 Autoqml: automatic generation and training of robust quantum-inspired classifiers by using evolutionary algorithms on grayscale images *Expert Syst. Appl.* **244** 122984
- [17] Yan F, Huang H, Pedrycz W and Hirota K 2024 Review of medical image processing using quantum-enabled algorithms *Artif. Intell. Rev.* **57** 300



- [18] Senokosov A, Sedykh A, Sagingalieva A, Kyriacou B and Melnikov A 2024 Quantum machine learning for image classification *Mach. Learn.: Sci. Technol.* **5** 015040
- [19] Hai V T, Viet N T, Urbaneja J, Linh N V, Tran L N and Ho L B 2024 Multi-target quantum compilation algorithm *Mach. Learn.: Sci. Technol.* **5** 045057
- [20] Pellow-Jarman R, Pillay A, Sinayskiy I and Petruccione F 2024 Hybrid genetic optimization for quantum feature map design *Quantum Mach. Intell.* **6** 45
- [21] Deb K, Pratap A, Agarwal S and Meyarivan T 2002 A fast and elitist multiobjective genetic algorithm: Nsga-ii *IEEE Trans. Evol. Comput.* **6** 182–97
- [22] Wang H 2025 Several fitness functions and entanglement gates in quantum kernel generation *Quantum Mach. Intell.* **7** 7
- [23] Martínez-Peña R, Soriano M C and Zambrini R 2024 Quantum fidelity kernel with a trapped-ion simulation platform *Phys. Rev. A* **109** 042612
- [24] Babu A, Ghatnekar S G, Saxena A and Mandal D 2025 Entanglement-enabled quantum kernels for enhanced feature mapping *APL Quantum* **2** 016116
- [25] Mohanty N, Behera B K and Ferrie C 2024 Solving the vehicle routing problem via quantum support vector machines *Quantum Mach. Intell.* **6** 34
- [26] Farooq O, Shahid M, Arshad S, Altaf A, Iqbal F, Vera Y A M, Flores M A L and Ashraf I 2024 An enhanced approach for predicting air pollution using quantum support vector machine *Sci. Rep.* **14** 19521
- [27] Shi C, Leoni G, Petrillo M, Gallardo A P and Wang H 2025 Compare similarities between DNA sequences using permutation-invariant quantum kernel (arXiv:2503.05465)
- [28] Glick J R, Gujarati T P, Corcoles A D, Kim Y, Kandala A, Gambetta J M and Temme K 2024 Covariant quantum kernels for data with group structure *Nat. Phys.* **20** 479–83
- [29] Thanasilp S, Wang S, Nghiem N A, Coles P J and Cerezo M 2023 Subtleties in the trainability of quantum machine learning models *Quantum Mach. Intell.* **5** 21
- [30] Peters E and Schuld M 2023 Generalization despite overfitting in quantum machine learning models *Quantum* **7** 1210
- [31] Jerbi S, Fiderer L J, Nautrup H P, Kübler J M, Briegel H J and Dunjko V 2023 Quantum machine learning beyond kernel methods *Nat. Commun.* **14** 517
- [32] Hubregtsen T, Wierichs D, Gil-Fuster E, Derks P-J H S, Faehrmann P K and Meyer J J 2022 Training quantum embedding kernels on near-term quantum computers *Phys. Rev. A* **106** 042431
- [33] Xu L, Zhang X-y, Li M and Shen S-q 2024 Quantum classifiers with a trainable kernel *Phys. Rev. Appl.* **21** 054056
- [34] Altares-López S, Ribeiro A and García-Ripoll J J 2021 Automatic design of quantum feature maps *Quantum Sci. Technol.* **6** 045015
- [35] Suzuki Y, Yano H, Gao Q, Uno S, Tanaka T, Akiyama M and Yamamoto N 2020 Analysis and synthesis of feature map for kernel-based quantum classifier *Quantum Mach. Intell.* **2** 9
- [36] Chen B-S and Chern J-L 2022 Generating quantum feature maps for SVM classifier (arXiv:2207.11449)
- [37] Gil-Fuster E, Eisert J and Dunjko V 2024 On the expressivity of embedding quantum kernels *Mach. Learn.: Sci. Technol.* **5** 025003
- [38] Moretti R, Giachero A, Radescu V and Grossi M 2025 Enhanced feature encoding and classification on distributed quantum hardware *Mach. Learn.: Sci. Technol.* **6** 015056
- [39] Acampora G, Chiatto A and Vitiello A 2023 Genetic algorithms as classical optimizer for the quantum approximate optimization algorithm *Appl. Soft Comput.* **142** 110296
- [40] Alhijawi B and Awajan A 2024 Genetic algorithms: theory, genetic operators, solutions and applications *Evol. Intell.* **17** 1245–56
- [41] Dong Y and Zhang J 2021 An improved hybrid quantum optimization algorithm for solving nonlinear equations *Quantum Inf. Process.* **20** 134
- [42] Wang F, Xie K, Han L, Han M and Wang Z 2023 Research on support vector machine optimization based on improved quantum genetic algorithm *Quantum Inf. Process.* **22** 380
- [43] Zhang Y, Zhao J, Jia Y and Shen X 2025 An improved adaptive quantum genetic algorithm as classical optimizer for the quantum approximate optimization algorithm on maxcut problem *Quantum Inf. Process.* **24** 222
- [44] Gen M and Lin L 2023 *Genetic Algorithms and Their Applications* (Springer) pp 635–74
- [45] Li J, Liu R and Wang R 2024 Handling dynamic capacitated vehicle routing problems based on adaptive genetic algorithm with elastic strategy *Swarm Evol. Comput.* **86** 101529
- [46] Hearst M, Dumais S, Osuna E, Platt J and Scholkopf B 1998 Support vector machines *IEEE Intell. Syst. Their Appl.* **13** 18–28
- [47] Noble W S 2006 What is a support vector machine? *Nat. Biotechnol.* **24** 1565–7
- [48] Incudini M, Martini F and Pierro A D 2025 Toward useful quantum kernels *Adv. Quantum Technol.* **8** 2300298
- [49] Egginger S, Sakhnenko A and Lorenz J M 2024 A hyperparameter study for quantum kernel methods *Quantum Mach. Intell.* **6** 44
- [50] Ding C, Wang S, Wang Y and Gao W 2025 Quantum machine learning for multiclass classification beyond kernel methods *Phys. Rev. A* **111** 062410
- [51] Schnabel J and Roth M 2025 Quantum kernel methods under scrutiny: a benchmarking study *Quantum Mach. Intell.* **7** 58
- [52] Javadi-Abhari A et al 2024 Quantum computing with qiskit (arXiv:2405.08810)
- [53] Kreplin D A, Willmann M, Schnabel J, Rapp F, Hagelūken M and Roth M 2025 sqlearn: a python library for quantum machine learning [focus: Quantum software and its engineering] *IEEE Softw.* **42** 65–72
- [54] Pedregosa F et al 2011 Scikit-learn: machine learning in python *J. Mach. Learn. Res.* **12** 2825–30
- [55] Abadi M et al 2016 TensorFlow: a system for Large-Scale machine learning *12th Usenix Symp. on Operating Systems Design and Implementation (Osdi 16)* (USENIX Association) pp 265–83
- [56] Kaynak C 1995 Methods of combining multiple classifiers and their applications to handwritten digit recognition *Master's Thesis Bogazici University Unpublished*
- [57] Xiao H, Rasul K and Vollgraf R 2017 Fashion-mnist: a novel image dataset for benchmarking machine learning algorithms (arXiv:1708.07747)
- [58] Aeberhard S and Forina M 1992 Wine (UCI Machine Learning Repository) (available at: <https://doi.org/10.24432/C5PC7J>)
- [59] Wolberg W, Mangasarian O, Street N and Street W 1993 Breast cancer wisconsin (Diagnostic) UCI Machine Learning Repository (available at: <https://doi.org/10.24432/C5DW2B>)
- [60] Dietterich T G 1998 Approximate statistical tests for comparing supervised classification learning algorithms *Neural Comput.* **10** 1895–923



- [61] Sohil F, Sohali M U and Shabbir J 2022 An introduction to statistical learning with applications in R *Stat. Theory Relat. Fields* **6** 87–87
- [62] Bravyi S, Sheldon S, Kandala A, McKay D C and Gambetta J M 2021 Mitigating measurement errors in multiqubit experiments *Phys. Rev. A* **103** 042605
- [63] Nation P D, Kang H, Sundaresan N and Gambetta J M 2021 Scalable mitigation of measurement errors on quantum computers *PRX Quantum* **2** 040326
- [64] Nguyen M D and Vu T H 2026 GA-QSVM (available at: <https://github.com/vutuanhai237/GA-QSVM>)